21-242 Recitation 12

Matrices — Q6. When is [T A] =A for every A?

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

For which $\mathcal{C}, \mathcal{B}$ can we say for all A that $[T_A]_{\mathcal{CB}} = A$?

Course-level solution

Here $T_A : \mathbb{F}^n \to \mathbb{F}^m$, so $\mathcal{B}$ is a basis of $\mathbb{F}^n$ and $\mathcal{C}$ is a basis of $\mathbb{F}^m$. Let P be the matrix with columns the $\mathcal{B}$-vectors (so P is the change-of-basis matrix from $\mathcal{B}$-coordinates to standard coordinates: $\vec{x} = P[\vec{x}]_{\mathcal{B}}$), and let Q be the matrix with columns the $\mathcal{C}$-vectors. Then

$$T_A(\vec{x}) = A\vec{x} \implies Q[T_A(\vec{x})]_{\mathcal{C}} = AP[\vec{x}]_{\mathcal{B}},$$

hence $[T_A]_{\mathcal{CB}} = Q^{-1}AP$.
We need $Q^{-1}AP = A$ for *every* $A \in \mathbb{F}^{m\times n}$. Taking $A = I$ (when $m = n$) already forces $Q^{-1}P = I$, so $P = Q$; then $Q^{-1}AQ = A$ for all A forces Q to commute with every matrix, hence Q is scalar, but $Q^{-1}AQ = A$ for all A in the rectangular case as well:
Take $A = E_{ij}$ (matrix units). Then $Q^{-1}E_{ij}P = E_{ij}$ for all i, j implies $P = I_n$ and $Q = I_m$.
Thus $[T_A]_{\mathcal{CB}} = A$ for all A if and only if $\mathcal{B}$ is the standard basis of $\mathbb{F}^n$ and $\mathcal{C}$ is the standard basis of $\mathbb{F}^m$.

Later-course solution (using ideas from the rest of the handout)

In one line: $[T_A]_{\mathcal{CB}} = Q^{-1}AP$, and the only pair of invertible matrices satisfying $Q^{-1}AP = A$ identically is $P = I$, $Q = I$. (If $m = n$ this is the statement that the only change of basis leaving every matrix numerically unchanged is the identity basis.)